

Responsive Resource-Efficient Continuous Learning for Video Analytics

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Introduction



Video analytics using DNNs is widely used in :

- Surveillance
- Traffic management
- Transportation

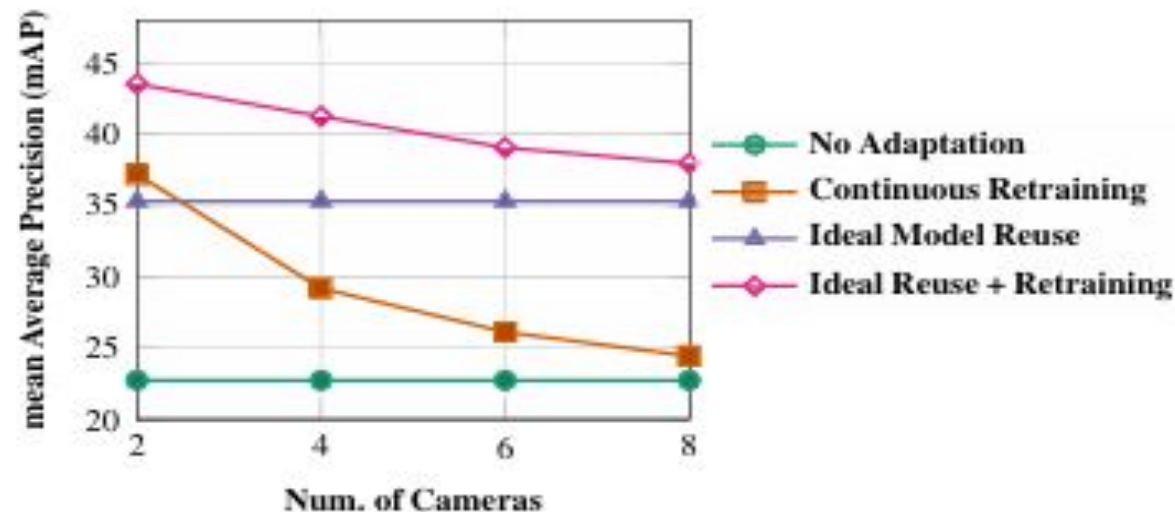
Limitations of existing approaches :

- Edge devices have limited GPU resources
- High compute cost
- Retraining delay causes accuracy drop
- Does not fully adapt to new scenes

Continuous Retraining for Video Analytics



- Generic DNNs are often too expensive to run for video analytics all the time.
- We observe that these systems need to spend 70%–90% of the overall compute resource on retraining their expert models.
- The idea is to use knowledge distillation to transfer the knowledge.



- RECL combines model reuse + online retraining efficiently
- We need to keep the cost of model reuse much lower.

1. Shared Model Zoo

Stores historical specialized expert models
Shared across multiple edge devices

2. Fast Model Selection

Quickly selects the best expert using a **lightweight gating network**

3. Dynamic GPU Scheduling

Prioritizes useful retraining jobs
Avoids wasting GPU on already-good models



RECL System Architecture



Controller runs on a **server or cloud machine**

Edge devices run:

- Real-time inference using lightweight expert models

Controller responsibilities:

- Select suitable models for edge devices
- Retrain models using sampled frames
- Maintain a shared model zoo



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- Time is divided into **model-update windows** (default: 30 seconds)
 - In each update window:
 - Edge device samples a small number of frames
 - Sampled frames are sent to the controller
 - Controller decides:
 - Whether to reuse an existing model
 - Or retrain a new model

RECL Implementation



Implemented RECL in **Python** using **PyTorch** for model training and inference

gRPC is used for communication between distributed services

System is built using **multiple microservices** for modularity and scalability

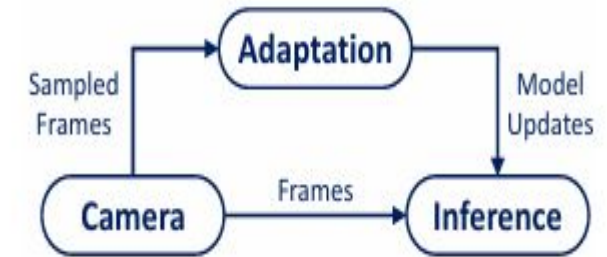
Microservices Architecture

- Camera streaming service:
 - Runs on each camera device
 - Sends subsampled video frames to the server
- Teacher labeler service:
 - Runs on the adaptation server
 - Labels frames using an accurate teacher model
- Runner service:
 - Coordinates all components across multiple video streams
 - Manages shared resources on the server

Key Results And Insights



- RECL outperforms all baselines on both tasks
- Improves:
 - Object detection mAP by **up to 9.0%**
 - Classification accuracy by **up to 7.4%**
- Baselines need **3.2× more compute** to match RECL accuracy



Responsiveness

- RECL adapts faster to scene changes
- Achieves target accuracy **91 seconds faster** than baselines

Impact of Model Reuse

- Major accuracy gains come from **reusing historical models**
- Compared to AMS and Ekya:
 - Up to **9.8% mAP** gain (detection)
 - Up to **10.7% accuracy** gain (classification)
- Model reuse significantly reduces retraining cost

Long-Term Behavior

- Compute overhead **decreases over time**
- As model zoo grows:
 - Model reuse increases
 - Cost of selector updates reduces
- Pruning keeps zoo compact with minimal accuracy loss

Conclusion



RECL Contributions

- First end-to-end system integrating:
 - Model reuse
 - Online model retraining
- Key components:
 - Fast and robust model selection
 - Shared model zoo across edge devices
 - Iterative, progress-aware training scheduler

Performance Benefits

- Higher accuracy than current systems
- Better resource efficiency at the same time

Thank You

A decorative graphic in the bottom right corner consisting of several light teal rectangular bars of varying lengths and orientations, creating a sense of movement or a stylized architectural element.